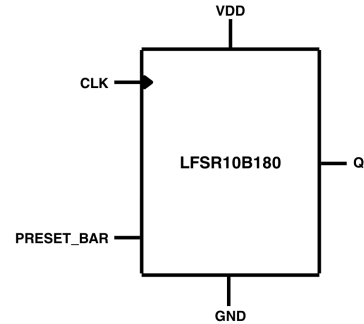


LFSR10B180

10-Bit Positive-Edge-Triggered Linear Feedback Shift Register with Asynchronous Preset

General Description

This device contains a full-custom, transistor-level 10-bit Linear Feedback Shift Register (LFSR) with a single serial output. The information on the internal sequential stages is shifted on the - positive-going edge of the clock pulse. The combinational feedback network utilizes a Galois Field polynomial of $x^{10} + x^7 + 1$ to generate a 1023-state Maximal Length Sequence (MLS). The data on the internal stages may be evaluated while the clock is LOW or HIGH without affecting the outputs as long as the data setup and hold times are not violated. A low logic level on the preset input will set or reset the internal latches to a deterministic hardware seed (1100000000) regardless of the logic levels of the other inputs.



Connection Information

Pin Name	I/O Type	Description
VDD	Power	Core supply voltage (1.8V nominal).
GND	Power	Circuit ground reference (0.0V).
CLK	Input	System Clock.
PRESET_BAR	Input	Asynchronous Active-Low Initialization.
Q	Output	1-bit Serial Output (Tap from stage 10).

Function Table

Input	Input	Output	Operating Mode
PRESET_BAR	CLK	Q	
L	X	L (Note 1)	Seed Injection
H	↑	PRBS	Sequence Generation
H	L	Q0	Hold / Inactive
H	H	Q0	Hold / Inactive

H = HIGH Logic Level

L = LOW Logic Level

X = Either LOW or HIGH Logic Level

↑ = Positive-going Transition

Q0 = output logic level of Q before the indicated input conditions were established

Note 1: During the active-low preset injection, the 10th stage is internally forced to a LOW logic level, resulting in Q = L

Absolute Maximum Ratings (Note 2)

Supply Voltage	1.8V
Input Voltage	1.8V

Note 2: The "Absolute Maximum Ratings" are those values beyond which the safety of the device cannot be guaranteed. The device should not be operated at these limits. The parametric values defined in the Electrical Characteristics tables are not guaranteed at the absolute maximum ratings. The "Recommended Operating Conditions" table will define the conditions for actual device operation.

Recommended Operating Conditions

Symbol	Parameter	Min	Nom	Max	Units
VCC	Supply Voltage	1.62	1.8	1.98	V
VIH	HIGH Level Input Voltage	1.2	--	1.8	V
VIL	LOW Level Input Voltage	0	--	0.6	V
TA	Temperature	0	25	-	°C

Electrical Characteristics

Symbol	Parameter	Conditions	Typ (Note 3)	Units
VOH	HIGH Level Output Voltage	VCC = Min, IOH = Max	1.812	V
VOL	LOW Level Output Voltage	VCC = Min, IOL = Max	0.012	V
I_dyn	Dynamic Current	Active switching at Max Frequency	136.7	µA
P_dyn	Dynamic Power	Active switching at Max Frequency	246.1	µW
Icc	Supply Current	VCC = Max, Idle state (Static)	10.2	nA

Note 3: All typicals are at VCC = 1.8V, TA = 25°C

Switching Characteristics

at VCC = 1.8V and TA = 25°C

Symbol	Parameter	From (Input) To (Output)	Typ	Units
fmax	Maximum Frequency	CLK to Datapath	1.75	GHz
tcq	Propagation Delay Time	Clock to Q	369.2	ps
tsu	Setup Time	Internal Feedback to Clock	78.5	ps
th	Hold Time	Internal Feedback to Clock	14.2	ps
tr	Rise Time (10% to 90%)	Clock to Q	84.3	ps
tf	Fall Time (90% to 10%)	Clock to Q	76.1	ps

Physical Dimensions

Macro optimized for SCL 180nm top-level integration

Parameter	Dimension	Units
Macro Width (X)	95	µm
Macro Height (Y)	80	µm
Total Silicon Area	7855.28	µm²

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